

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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Sales Item Management System - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Sales Item Management System is a client-side front-end application designed for browsing and managing a product catalog. It implements a full CRUD lifecycle for items, featuring a responsive gallery, dynamic filtering, and an administrative management interface. The system follows a client-side data pattern, focusing on interactive UI components like modals for detailed item views.

1.2 Maturity Assessment

The project is currently in **REFINEMENT**. While functional requirements and entity mappings are complete, there are critical structural gaps regarding high-level business goals and explicit scope boundaries. Most notably, the persistence strategy remains an unresolved uncertainty, which prevents the finalization of the data architecture.

1.3 Technical Stack

- HTML
- CSS
- JS

1.4 Architectural Constraints

- Responsive layout compatible with mobile, tablet, and desktop.
- Price validation: must be a positive number.
- Input validation: prevent submission of empty item names.
- UI Performance: Item addition must be completed in under 30 seconds.
- Visual Consistency: 100% of catalog items must follow a consistent layout.
- Empty State: Mandatory "No items found" message when catalog is empty.

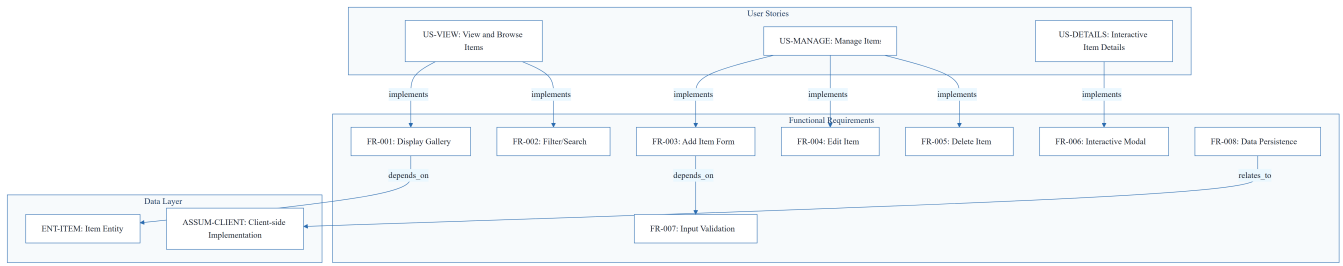
1.5 Critical Dependencies

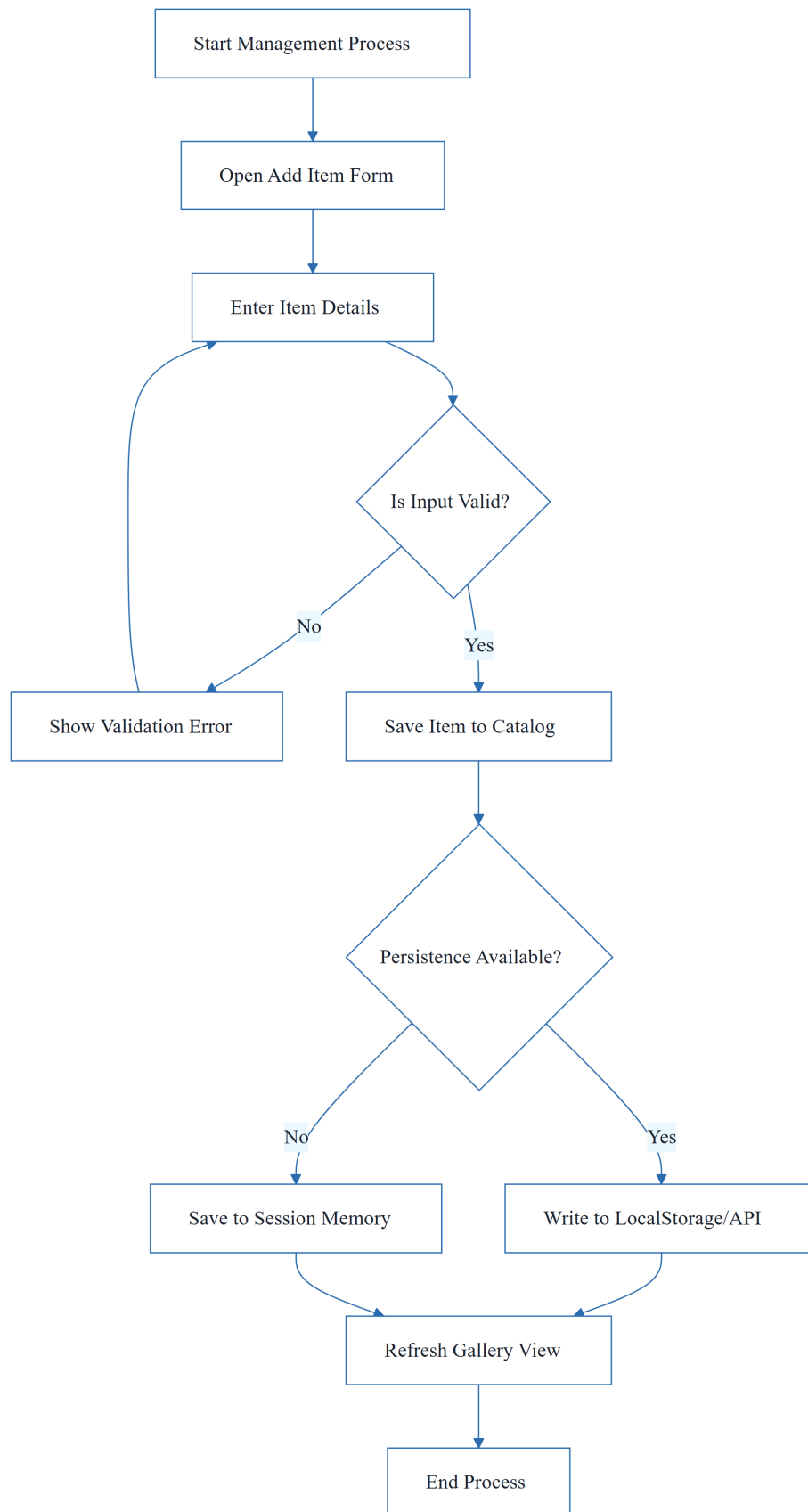
- **Persistence Layer:** Decision required between LocalStorage or Mock API/JSON for data durability across reloads.
- **Entity Integrity:** Item entity requires a Unique ID for CRUD operations (Update/Delete).
- **Client-side Runtime:** Browser environment supporting modern JS for dynamic DOM manipulation.

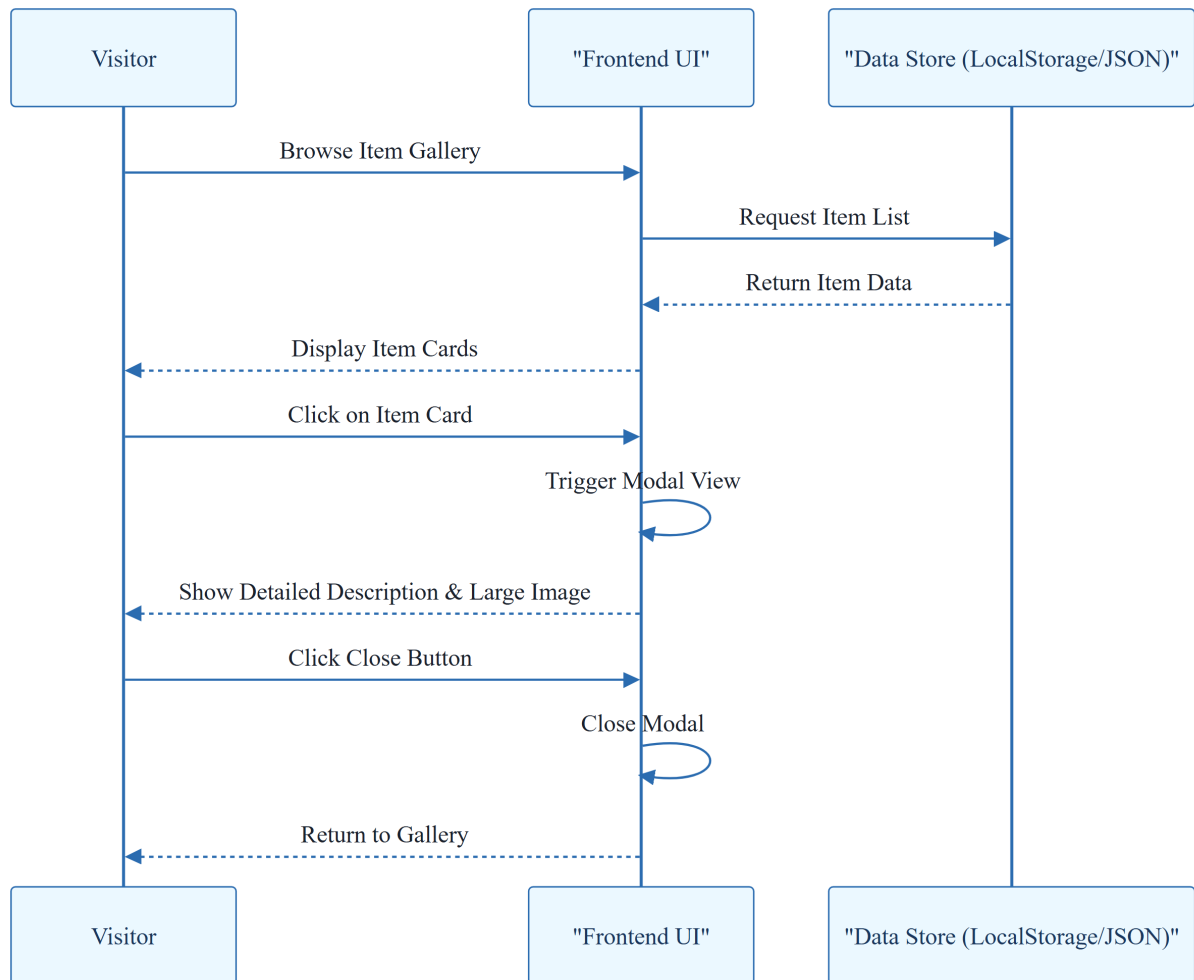
- **Asset Pipeline:** ImageURL dependency for product gallery rendering.

2. Architecture Workflows

ENT-ITEM		
string	id	PK
string	name	
string	description	
number	price	
string	category	
string	imageURL	





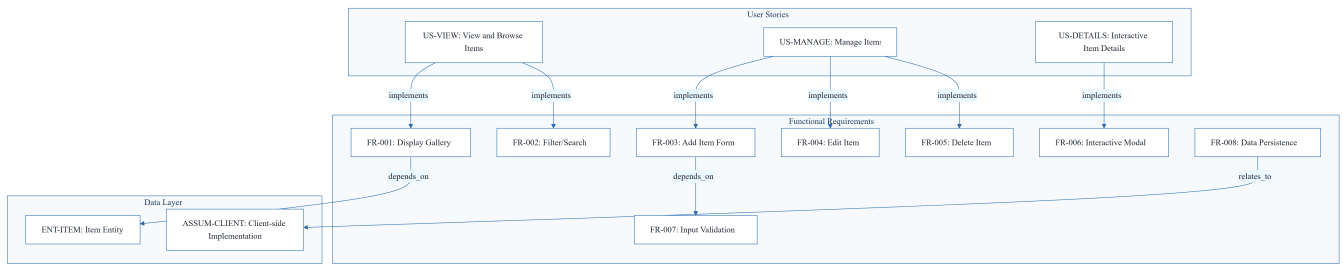


& Visual Diagrams

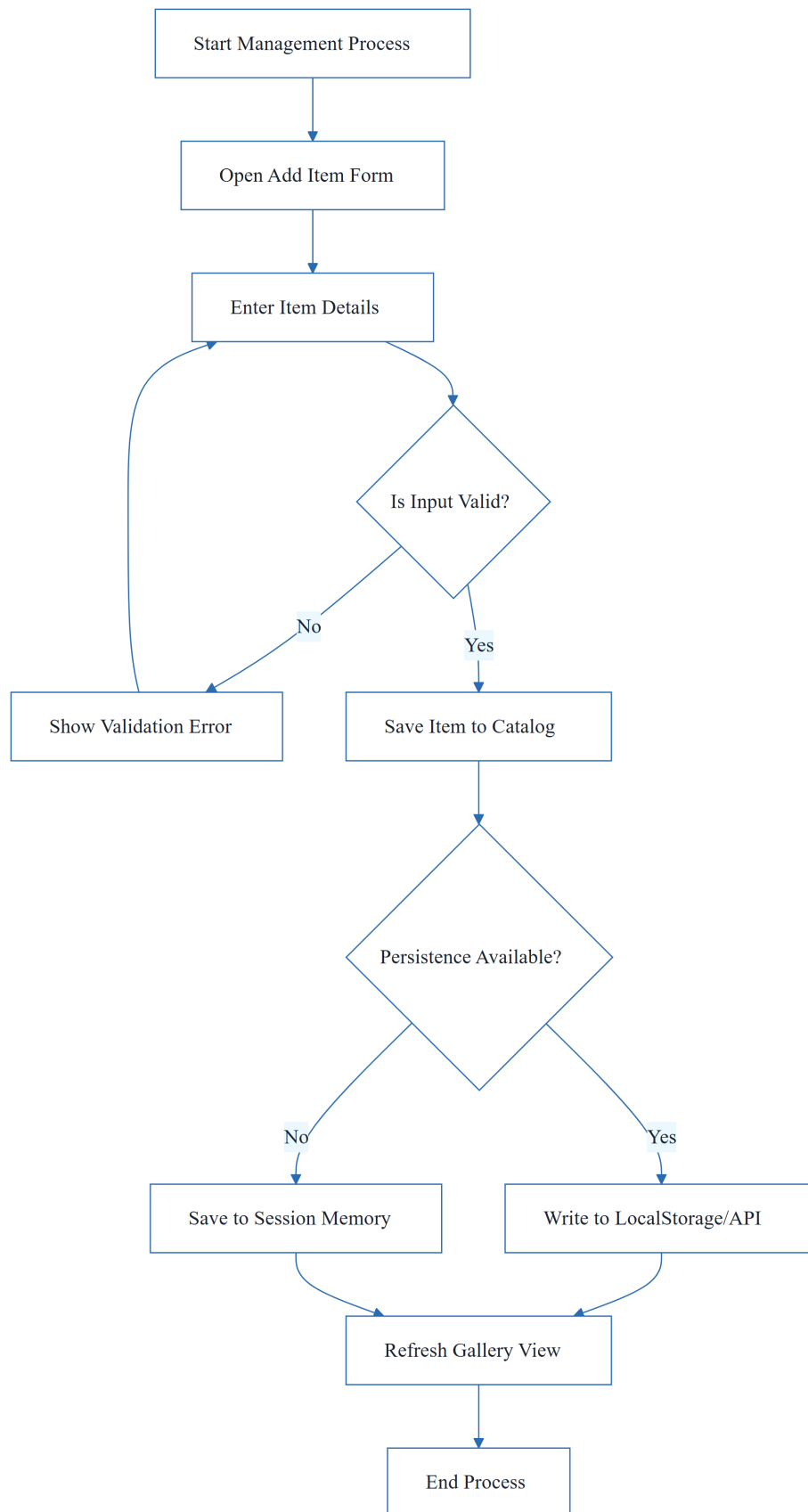
2.1 Sales Item Management Data Model

ENT-ITEM		
string	id	PK
string	name	
string	description	
number	price	
string	category	
string	imageURL	

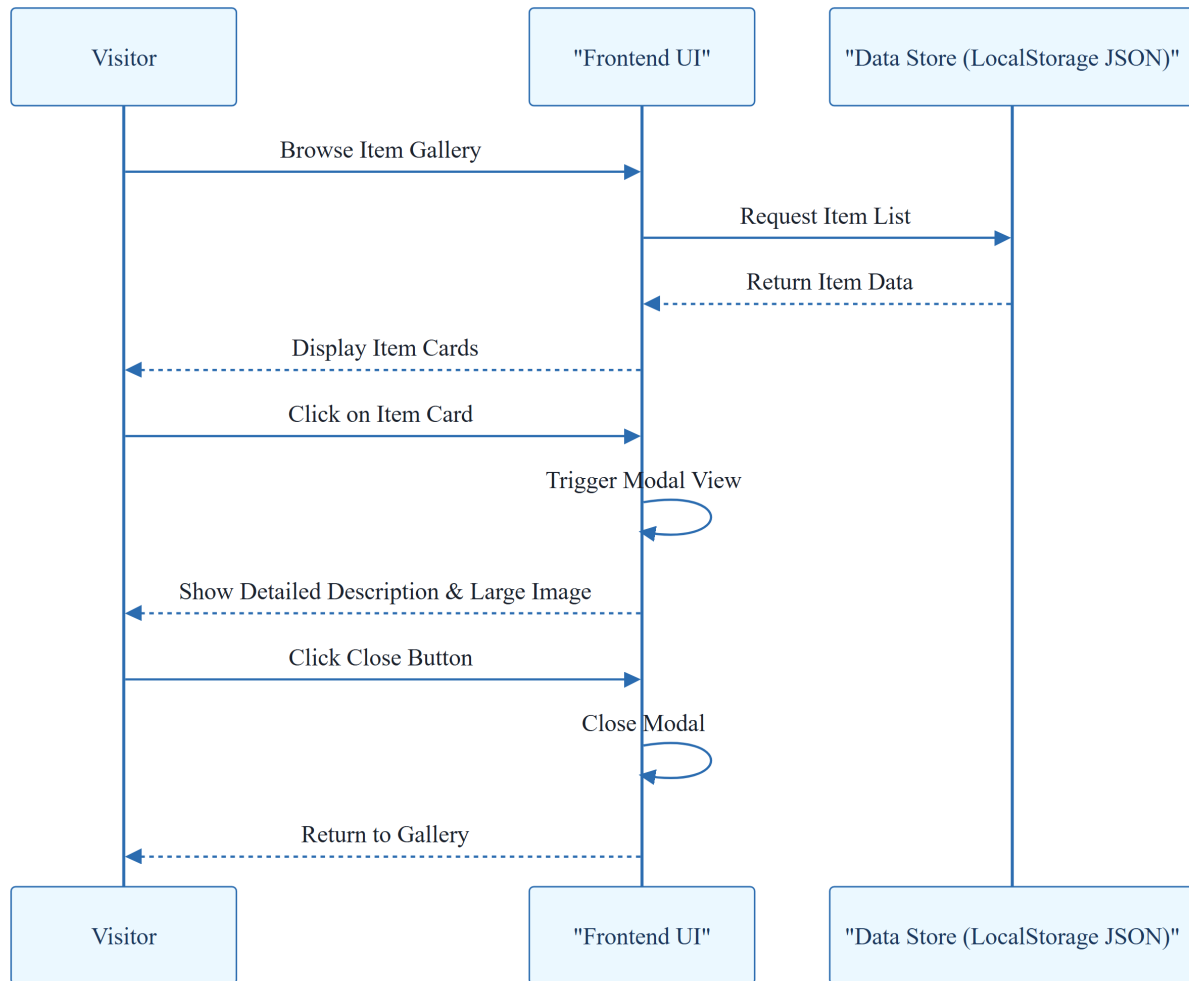
2.2 Requirements Traceability Matrix



2.3 Item Management Workflow



2.4 Item Interaction Sequence



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Type	Description	Source / Relation
US-VIEW	User Story	As a visitor, I want to see a list of items available for sale so that I can find products that interest me.	P1
US-MANAGE	User Story	As an administrator, I want to add, edit, and delete items from the catalog so that the inventory remains up-to-date.	P1
US-DETAILS	User Story	As a visitor, I want to click on an item to see more details or a larger image so that I can make an informed purchase decision.	P2

ID	Type	Description	Source / Relation
FR-001	Functional	System MUST display a gallery of items for sale.	US-VIEW
FR-002	Functional	System MUST allow users to filter items by category or search by name.	US-VIEW
FR-003	Functional	System MUST provide a form to add new items to the catalog.	US-MANAGE
FR-004	Functional	System MUST allow editing of existing item details.	US-MANAGE
FR-005	Functional	System MUST allow deletion of items from the catalog.	US-MANAGE
FR-006	Functional	System MUST display item details in an interactive modal.	US-DETAILS
FR-007	Functional	System MUST validate form inputs (e.g., price must be a positive number).	FR-003
FR-008	Functional	System MUST persist item data across page reloads.	ASSUM-CLIENT
REQ-EMPTY	Functional	The system should display a 'No items found' message when no items are available.	Edge Cases
NFR-RESPONSIVE	Non-Func	The interface is fully responsive and usable on mobile, tablet, and desktop.	Success Criteria
ENT-ITEM	Entity	Item: ID, Name, Description, Price, Category, ImageURL	FR-001
ASSUM-CLIENT	Assumption	The project is a client-side implementation (HTML/CSS/JS).	FR-008
CONSTR-DESIGN	Constraint	Modern design implies a clean, minimalist aesthetic with a responsive layout.	Assumptions

3.2 Security Rules

- **Input Validation:** All form submissions must be validated on the client side to prevent invalid data (e.g., negative prices) from entering the data store.
- **Administrative Access:** Management functions (Add/Edit/Delete) are handled via a simple admin interface on the same site (Note: No authentication layer specified in current scope).

3.3 Data Models

Entity: Item (ENT-ITEM)

- `id`: Unique identifier (String)
- `name`: Product name (String)
- `description`: Detailed product description (String)
- `price`: Unit price (Number)
- `category`: Product category (String)
- `imageUrl`: Path or URL to the product image (String)

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Goals & Objectives	HIGH	Define the high-level business goals and the primary objective of the Sales Item Management system.
Scope & Out-of-Scope	MEDIUM	Explicitly state what the system will NOT do (e.g., no payment gateway, no user authentication).
Open Questions & Uncertainties	LOW	Create a dedicated section for the persistence strategy (LocalStorage vs API).

4.2 Remediation & Workflow

The primary blocker for technical finalization is the **Persistence Strategy**. The development workflow must prioritize the decision between `LocalStorage` (for simple client-side persistence) and a `Mock API/JSON` (for simulated backend integration) before implementing the `FR-008` requirement.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	FR-008	A potential external interface for data exchange and remote storage retrieval.
CRUD	TECHNICAL_STACK	US-MANAGE	The four foundational persistent storage mutation primitives.

Term	Category	Context Anchor	Project Definition
CSS	TECHNICAL_STACK	ASSUM-CLIENT	The styling language used to implement the minimalist and responsive visual layout.
Empty Catalog	BUSINESS_DOMAIN	REQ-EMPTY	A state where no available products exist, triggering a specific notification message.
Fixed-Point Numeric Constraint	TECHNICAL_STACK	FR-007	A validation rule ensuring monetary values are positive and formatted correctly.
HTML	TECHNICAL_STACK	ASSUM-CLIENT	The structural markup used to define the page content and gallery layout.
ID	BUSINESS_DOMAIN	ENT-ITEM	A unique alphanumeric token assigned to each product for precise tracking.
Invalid Input	BUSINESS_DOMAIN	Edge Cases	Data entries that violate system rules, such as negative pricing or missing names.
Item	BUSINESS_DOMAIN	ENT-ITEM	The primary entity containing a name, description, price, category, and image reference.
JS	TECHNICAL_STACK	ASSUM-CLIENT	The scripting language enabling dynamic updates and interactive modal behavior.
JSON	TECHNICAL_STACK	FR-008	A lightweight data-interchange format used for mock data or storage serialization.
LocalStorage	TECHNICAL_STACK	FR-008	The browser-based key-value storage mechanism for maintaining data across sessions.
Measurable	TECHNICAL_STACK	Success Criteria	A quality metric defined by a specific time threshold for completing a task.
Persistence	TECHNICAL_STACK	FR-008	The capability to retain state and data after a browser refresh.

Term	Category	Context Anchor	Project Definition
Technology-Agnostic	TECHNICAL_STACK	NFR-RESPONSIVE	A design approach ensuring accessibility across various device types and screen sizes.
User-Focused	BUSINESS_DOMAIN	Success Criteria	A requirement for visual consistency across all displayed product cards.
Verifiable	TECHNICAL_STACK	Success Criteria	The state where all core data mutation operations are confirmed as functional.